

CNSS Gap Analysis: Data Tagging Policy, Standards and Implementation in Support of a Zero Trust Architecture, 24 March 2021

1 Introduction

The Committee on National Security Systems (CNSS) convened a Data Tagging Tiger Team in October, 2020, to examine existing DoD, IC and Federal policy and data tagging standards and identify gaps and conflicts that would prevent implementation of a Zero Trust architecture across the NSS Enterprise. The following organizations participated in the CNSS Data-Tagging Tiger Team from October 2020 to the present:

National Security Agency – Tri- chair

Central Intelligence Agency - Tri- chair

Defense Information Systems Agency- Tri-chair

Department of Defense

Department of Energy

Department of State

Office of the Director of National Intelligence

United States Air Force

1.1 Background

“The purpose for a Zero Trust Architecture (ZTA) is to protect data.”¹

“Zero Trust is a cybersecurity strategy and framework that embeds security throughout the architecture for the purpose of stopping data breaches. This data centric security model eliminates the idea of trusted or untrusted networks, devices, personas, or processes; and shifts to multi-attribute based confidence levels that enable authentication and authorization policies under the concept of least privileged access. Zero Trust requires a paradigm shift, from a legacy defense in depth security model that is hierarchical and transport focused, to a data centric model optimized for the enterprise. We must re-think how we utilize existing infrastructure to implement security by design in a simpler and more efficient way while enabling unimpeded operations.”²

The shift to a data centric model requires a huge paradigm shift. Data centric refers to an architecture where data is the primary and permanent asset, and applications come and go. In the traditional net-centric security model, securing the boundaries of a system equated to securing data contained on the

¹ American Council for Technology-Industry Advisory Council (ACT-IAC), Zero Trust Cybersecurity Trends, 18 April, 2019.

² DoD Digital Modernization Strategy, DoD Information Resource management Strategic Plan FY19-23, cleared for open publication on 12 July 2019.

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system. In the data centric model, supporting information sharing, data is expected to move between systems and networks in support of mission tasks.

Data tagging, or, more specifically, metadata tagging, is a standard technical method of defining, indexing and working with large volumes of data, and is the documented technical approach defined in to the Strategic Implementation Plan for the National Strategy for Information Sharing and Safeguarding. The Data Management Body of Knowledge states that metadata “contributes to the ability to process, maintain, integrate, secure audit, and govern other data.”

At the enterprise level, the *2012 National Strategy for Information Sharing and Safeguarding* lists adopting “metadata standards to facilitate federated discovery, access , correlation and monitoring across Federal networks and security domains,”³ as the third of its top five Priority Objectives.

Access control is traditionally implemented by the enforcement of a policy decision at a policy enforcement point. For network level security, this policy enforcement point is located at the network boundary. Once approved, a person or non-person entity, also known as the subject, has access to all resources on the network, including all applications and data, generically called resource objects. This model, referred to as boundary defense, castle-and-moat, or chocolate bon-bon (hard shell, soft, vulnerable center), assumes that anyone within the boundary is trustworthy and should have access to resources residing there.

The “Zero Trust Architecture (ZTA) moves away from the perimeter approach and instead assumes, as the name would suggest, zero trust in the network itself... ZTA operates on a “least-privilege access” model by only granting users and devices access to the applications, services and data that are absolutely necessary for their role within an organization.”⁴

NIST 800-207 provides an operative definition of zero trust and zero trust architecture is as follows:

Zero trust (ZT) provides a collection of concepts and ideas designed to minimize uncertainty in enforcing accurate, least privilege per-request access decisions in information systems and services in the face of a network viewed as compromised. *Zero trust architecture (ZTA)* is an enterprise’s cybersecurity plan that utilizes zero trust concepts and encompasses component relationships, workflow planning, and access policies. Therefore, a zero trust enterprise is the network infrastructure (physical and virtual) and operational policies that are in place for an enterprise as a product of a zero trust architecture plan.⁵

The goal of ZT is to *prevent unauthorized access to data applications, assets, and services* coupled with making the *access control enforcement as granular as possible*.

³ National Strategy for Information Sharing and Safeguarding, December 2012, https://obamawhitehouse.archives.gov/sites/default/files/docs/2012sharingstrategy_1.pdf

⁴ American Council for Technology-Industry Advisory Council (ACT-IAC), Zero Trust Cybersecurity Trends, 18 April, 2019.

⁵ NIST SP 800-207, Zero Trust Architecture, 11 Aug 2020. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-207.pdf>

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This definition of ZTA strongly aligns with an attribute based access control (ABAC) implementation. ABAC is a model that controls access based on attributes assigned to the subject, object and environment, and supports dynamic access control, where the subject identity is not specifically known prior to the access request. The access policy decision is rendered based on subject authorization attributes compared to the object access control attributes contained in metadata statements linked to the data objects dynamically, at the time of the access request. For data objects, marking with control attributes is also known as data (or metadata) tagging. Metadata is data about data.

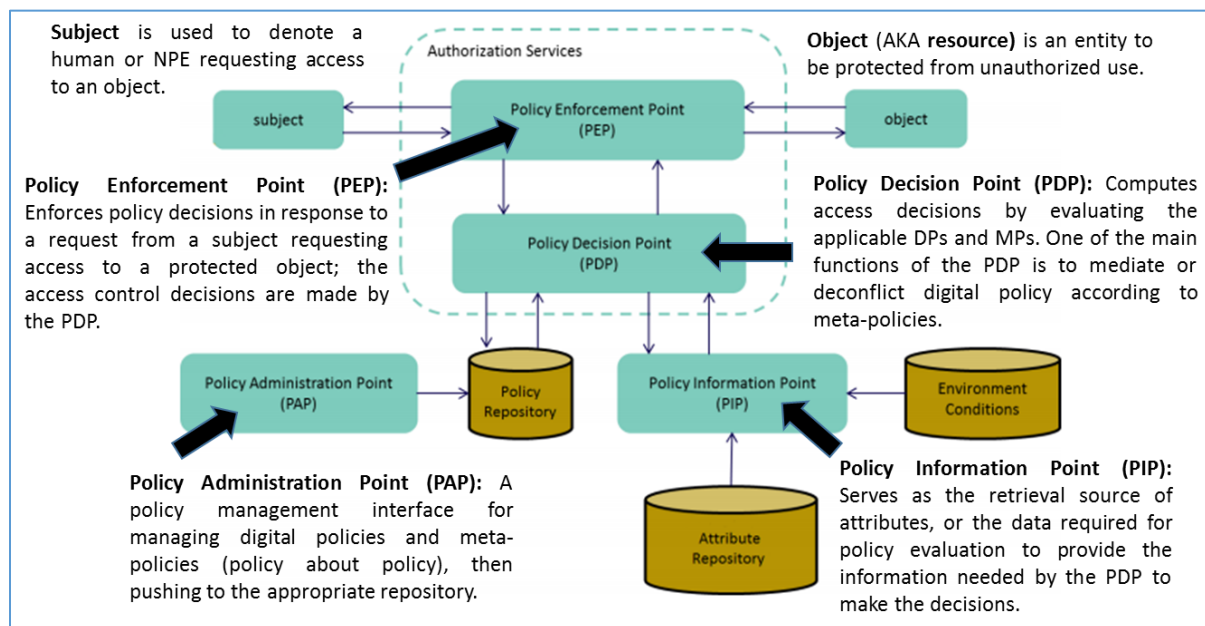


Figure 1. Access Control Mechanism Functional Points, From NIST SP 800-162, Guide to ABAC Definition and Considerations

Figure 1, modified from a graphic in NIST SP 800-162, Guide to ABAC Definition and Considerations⁶, outlines the functional points of access control.

⁶ NIST SP 800-162, Guide to ABAC Definition and Considerations, January 2014.
<https://doi.org/10.6028/NIST.SP.800-162>

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1.2 Purpose

The purpose of this document is to examine existing DoD, IC and Federal policy and data tagging standards and identify gaps and conflicts that would prevent implementation of a Zero Trust architecture across National Security Systems (NSS). The analysis will list pertinent policy documents identify appropriate standards and specifications to meet Zero Trust framework implementation requirements. It will review information based on system users responses to a simple list of questions, then analyze and provide recommendations for the CNSS way ahead.

1.3 Scope

Networks and systems investigated were identified based on the definition of National Security Systems, excerpted from NIST SP 800-59, Guideline for Identifying an Information System as a National Security System⁷, based on the definition provided in law (44 U.S.C. 3542(b)(2), which was established by FISMA, Title III, Public Law 107347, December 17, 2002). Within those systems, the scope of this review was limited to policy and standards related to data sharing and securing, as defined in the most recent National Strategy for Information Sharing and Safeguarding⁸. The companion document, the Strategic Implementation Plan for the National Strategy for Information Sharing and Safeguarding⁹, ranks data tagging as the third priority objective, after governance and sharing agreements, and directs agencies to:

Adopt metadata standards to facilitate federated discovery, access, correlation, and monitoring across Federal networks and security domains.

Recent updates to the Presidential and Federal Records Act¹⁰, and guidance from the Information Sharing Environment Priority Objective 3, Data Tagging Functional Requirements¹¹, drove the inclusion of metadata tagging in support of digital records management. Therefore, this team's review looked at data tagging for sharing and safeguarding with respect to discovery, access control, correlation and analysis, monitoring and federal records management.

1.4 Approach

A list of DoD, IC and Federal policy documents was developed and provided to the CNSS Data Tagging Tiger Team members for review. The Tiger Team developed a short questionnaire that was sent by the CNSS secretariat to CNSS members, with a request to forward for broad distribution. The questionnaire

⁷ Barker, William, National Institute for Standards, Guideline for Identifying an Information System as a National Security System, August 2003. <https://csrc.nist.gov/publications/detail/sp/800-59/final>

⁸ National Strategy for Information Sharing and Safeguarding, December 2012, https://obamawhitehouse.archives.gov/sites/default/files/docs/2012sharingstrategy_1.pdf

⁹ Information Sharing Environment (ISE) Strategic Implementation Plan for the National Strategy for Information Sharing and Safeguarding, December 2013. <https://www.hsdl.org/?view&did=748441>

¹⁰ Presidential and Federal Records Act Amendments of 2014, 26 November 2014, <https://www.congress.gov/bill/113th-congress/house-bill/1233>

¹¹ Information Sharing Environment Priority Object 3, Data Tagging Functional Requirements, December 2014, <https://stage.dni.akadns.net/files/ISE/documents/DocumentLibrary/PO3-Data-Tagging-Functional-Requirements.pdf>

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was sent out in early December, with a two week suspense. Due to the winter holidays, responses were accepted through the second week in January 2021.

A total of thirteen agencies/organizations responded, including three intelligence agencies (CIA, NSA, and ODNI), The Department of the Air Force Chief Data Officer, the Joint Artificial Intelligence Center (JAIC), four sub-organizations within USCYBERCOMMAND, USSOUTHCOM, the intelligence directorate at USCENTCOM, the USCENTCOM Joint Interoperability through Data Centricity (JI DC) Test organization, and a mission owner in the Cyber Mission area.

The questionnaire, included as Appendix A, was intended to gauge the knowledge of policy and standards and an understanding of the areas of discovery, access control, correlation and analysis, monitoring and federal records management, implementation and use of data tagging across National Security Systems. Thanks to Joint Staff J6, the questionnaire was forwarded as a tasking to all Combatant Commands, which yielded what the tiger team considered an exceptionally strong response from that sector.

1.5 Document Organization

This document is organized to include the requested policy review, information on data standards, analysis and findings from the questionnaire, and recommendations to the CNSS Committee.

2 Policy Review

As early as 2007, CNSS Policy guidance stated that, *“Information must be consistently and appropriately labeled with special handling instructions—metadata standards are needed.”*¹² The *Strategic Implementation Plan for the National Strategy for Information Sharing and Safeguarding* placed data tagging above the implementation of the Federal Identity, Credential, and Access Management (FICAM) Roadmap in its list of priority objectives.¹³

All of the documents included in the review (See Appendix B for list and links) reference the need for data and metadata standards in support of National Strategic objectives for secure data sharing.

3 Standards and Specifications Review

The document, *“Program Manager, Information Sharing Environment Data Tagging Functional Requirements,”*¹⁴ identifies five capability areas across nearly all Departments and Agencies that are essential to information sharing and safeguarding: Query and Discovery, Access Control, Correlation, Audit and Records Management. Within these areas, the DoD and IC maintain and mandate specifications and standards for data tagging which cover a majority of CNSS systems. These specifications and standards are governed by the Joint Enterprise Standards Committee (JESC).

¹² CNSS-007-07, Policy Development Guidance to Address Information Sharing for National Security Systems, 2007.

¹³ Strategic Implementation Plan for the National Strategy for Information Sharing and Safeguarding, December 2013. <https://www.hsd.org/?view&did=748441>

¹⁴ Program Manager, Information Sharing Environment Data Tagging Functional Requirements, a <https://www.dni.gov/files/ISE/documents/DocumentLibrary/PO3-Data-Tagging-Functional-Requirements.pdf>

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The JESC was chartered in March of 2013, by the DoD and IC, to govern information technology standards, and to establish, update, publish, maintain, and make accessible the baseline set of enterprise standards, profiles, and specifications to authorized parties.¹⁵

JESC mandated data standards, as listed in the DoD Information Technology Standards Registry (DISR), for sharing and safeguarding with respect to discovery, access control, correlation and analysis, monitoring and federal records management were used as a baseline to analyze the CNSS community's understanding of standards and specifications use. It was expected that those within the DoD and IC using metadata standards would be using DISR mandated standards. The specific standards list used as a baseline are given in Appendix C.

4 Analysis and Findings

Thirteen responses were received to the CNSS Data Tagging Tiger Team Questionnaire. The respondents spanned the DoD, and IC and included: CIA, Department of the Air Force, Joint Artificial Intelligence Center (JAIC), NSA, ODNI, USCENTCOM, USCYBERCOM J010, USCYBERCOM J2, USCYBERCOM J3, USCYBERCOM/J070 Information Defense Division, USEUCOM, and USSOUTHCOM. Their answers to the questionnaire provided a breadth of response, from responders aware of and using JESC mandated standards to organizations using commercial (potentially proprietary) standards, to organizations developing their own standards. There were four areas (see below) that we analyzed from the responses.

4.1 Disparity of Maturity

(U) The most mature organizations tend to be within the IC, where Agency Data Management is implemented through a Chief Data Officer's (CDO) Council, the IC and each Agency's Data Strategy and executed by the Agency Chief Data Officer through organizational directorate Data Officers, Data Stewards, and other Agency Governance Officers to ensure that information is tagged and discoverable.

(U) At the other end of the spectrum, there are organizations where no data tagging policy is enforced, or there is a gap between intelligence data and other, classified data, under DoD guidance.

4.2 Lack of Data Literacy

(U) The majority of respondents from the DoD stated that they were aware of the DoD policies, but unaware of, or not enforcing either the policy or the DISR mandated data standards. Several stated that they used data standards for records management, but not for information sharing. Many appear to believe that only USCYBERCOM data has any cybersecurity implications, and lack understanding that all data resources should implement some type of access control to match system users to authorized resources, including data.

(U) One respondent clearly stated the issue: "Many users/producers don't understand tagging, its function or need."

¹⁵ Joint Enterprise Standards Committee Charter, May 2013, <<https://gwg.nga.mil/documents/JESC%20Charter%20Signed%202013%2003%2028.pdf>>

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4.3 Lack of Resources

(U) Outside of IC respondents, most users secure their data in MS SharePoint repositories using SharePoint access control list functions. No effort is made to make data visible and accessible to authorized users outside of the immediate organization.

4.4 Lack of cohesive approach and/or governance of identity, cybersecurity, data tagging and related areas

(U) Too many of the respondents believed that their command level organization needed to create specific policy regarding data tagging and access control. Enterprise level mandates are not clearly defined and promulgated to mission level users, nor to acquisition and test organizations, so non-conforming solutions continue to flow to the end users.

5 Recommendations

(U) Federal data strategy clearly states that data is a federal asset, and that it should be made available to all authorized users. Zero Trust principles require an organization to know who is on their systems, and what authorized resource objects, systems and data, that subject is accessing. In implementation, this translates to strong digital identity based access control paired with strong resource object marking. In the case of data, this means clear metadata tagging for access control.

5.1 Provide Federal-level guidance for data tagging which articulates the driver for and ideally provides an incentive to participate in zero-trust enterprise

- a) (U) Develop a CNSS Data Tagging Directive, developed to parallel the guidance in CNSSD-507, National Directive for Identity, Credential, and Access Management to assist organizations in establishing formal programs to support Data Tagging efforts.
- b) (U) Develop a specific CNSS Data Tagging Instruction. In some categories, such as crypto keying, the CNSS Instruction would be the definitive Federal guidance for NSS to assist organizations in technical best practices.
- c) (U) Influence the creation of NIST guidance for data tagging.

5.2 Drive Zero Trust based implementation throughout organization

(U) Existing policy, guidance, and standards are heavily focused on sharing of information, which can be addressed by solutions short of zero-trust such as border security approaches. As the CNSS community moves to cloud based infrastructure, it can no longer only depend on border security. The boundary of the secure zone must move closer to the level of protected resource, the software and data. Said another way, the “Protect Surface” must decrease to the data itself. Provide strong technical guidance which clearly identifies when boundary security is not enough.

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5.3 Provide a framework through CNSSI 1253 compliance

(U) CNSS can provide a recommended Zero Trust framework to facilitate coordination across CIO, CDO, and CISO offices within organizations. This could be accomplished by including zero trust principles in overlays for CNSSI 1253, adding a Zero Trust Framework to CNSSP 21, or other mechanism. In some organizations, responsibility for the various pieces needed for a zero trust environment are scattered across various components and efforts are not as coordinated as is needed. A CNSS Zero Trust Framework provides clear, high- level direction/guidance to facilitate common direction and data – tagging approaches.

Appendix A. Data Tagging Questionnaire and Responses.

Q1. Are there any policies or regulations mandated for data tagging to facilitate information sharing within your organization or between other Department/Agencies? (Yes or No)

Yes – 4 / No - 7

Q2. How are mandated policies or regulations enforced (explain)? 8 respondents stated N/A.

Other responses to question 2:

A 2-1: (U) Agency Data Management is implemented through the Agency Data Strategy and executed by the Chief Data Officer (CDO) through organizational directorate Data Officers, Data Stewards, and other Agency Governance Officers to ensure that information is tagged and discoverable.

(U) Policies and regulations concerning data controls are enforced via multiple avenues, depending on the system/process in question. These include:

- (U) Evaluation as part of the Assessment and Authorization (A&A) process
 - (U) Evaluation as part of the Data Steward approval process
 - (U) Evaluation as part of additional certifications, such as the CDO (Chief Data Officer's) Data Repository program
 - (U) Reliance on Technical Specifications as the most concrete form of IC policies: to ensure compliance with policy, to ensure adherence to policy, for validation methods against policy. Primarily, the ODNI Intelligence Community Technical Specification XML Data Encoding Specifications (for Enterprise Data Header, for Information Resource Metadata, for Information Security Markings, for Need-To-Know Metadata, for Intelligence Publications).
 - (U) Governance concepts including Working Groups and Subcommittees for technical and implementation guidance (such as the Mission Data Subcommittee and the Access Control Subcommittee); and a Data Governance Council for escalation of issues for resolution, prioritization, and approvals.
 - (U) Ongoing review and recommendation/approval of technical standards/specifications IAW IC specifications.
 - (U) Tagging requirements specified within agency regulations (e.g., Regulations for EO12333 and PPD-28 tagging compliance).
 - (U) Mandated use of approved terms from the IC Data Management Lexicon.
 - (U) Automated audits (requirements such as for certain queries).
 - (U) Locally developed specifications and services (e.g., MINT).
 - (U) Mandated use of standard data cataloging services and standard enterprise access control services.
 - (U) Enterprise standards for certified data repositories. Ensures standard tags are used in a data lake and data is not ingested unless properly tagged. This provides data lake interoperability and enforces proper data object tagging.
- (U) Additionally, there are no hard mandates for data tagging of high-volume data. There are consensus standards that are often used when data is shared between agencies or when data is used in shared exploitation tools and platforms. These consensus standards are sometimes shared across organizational boundaries, but are not officially mandated through policy.

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Examples would be a standardized format for SIGINT data and the person model used in Identity Intelligence mission.

A 2-2. (U) The policies are enforced through access controls. Only personnel with the clearance and need to know can access shared data.

A 2-3. (U) Policies and regulations aren't enforced. However, we are implementing weekly announcements and conducting SAVs and providing results to leadership to ensure that J-Dirs are aware of their RM responsibilities which includes data tagging. Also, we will begin weekly RM trainings that are open to the Command.

A2-4 (U) Finished intelligence products are tagged, at least minimally, using the ICPUBS schema required to include the products in national databases. This metadata is reviewed by the directorate's dissemination element prior to publication on disparate networks.

A 2-5 (U) mandated data tagging policies are enforced through automations. Within the environment all uploaded/shared documents are required to be in the Trusted Data Format (TDF) or they are rejected by the software.

Q3. Is data tagging mandated for information sharing within your department/agency or organization? (Yes or No)

Yes – 4 / No – 6 / Sometimes – 2

Comments:

1. (U) Yes, mandated in the sense of broad policy pronouncements that tagging must be done using Agency and IC metadata standards. Agency systems (through A&A), ensure the evaluation and implementation of security controls which speak to tagging and/or marking data (unclassified or classified) to ensure control around Data Access, Need to Know (NTK), and Data Dissemination; both internally and externally.
(U) We do not have published mandates for metadata specifications required for every data type. This is especially true for semantic tagging and any tagging that results from conformance to specific data models.
2. (U) No. Each mission initiative (joint warfare, information warfare, human and disaster relief, preventive maintenance, health, and intelligent business automation) has a different domain but often overlapping data requirements. Since they run independently, it's hard to enforce a standard across all groups. Centralizing the data within the JCF would prevent redundancy and improve multi modal analysis
3. (U//FOUO) No. Standing: It seems that the Cyber Security Community (CSC), consisting of CSSPs, U.S. Cyber Command (USCYBERCOM) and service component cyber operations, Cyber National Mission Forces, as well as the numerous DoD cleared defense contractor service providers are NOT considered part of the NSS. We would argue that in multi-domain warfare that cyber is now an important element of NSS and likely a weapon system / defensive system

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as well.

Policy: The Cyber Security Community (CSC) continues to be hampered by legacy policies, outdated concepts of networking, outdated understanding of data ownership and outright prohibitions on any data sharing. The lack of a useful classification guide for such tagged data will also be required.

Data tagging for the purposes identified here could improve the state, given the above limitations and provide a starting point for identifying WHAT data are worthy to share, WHICH tags to apply during collection, and WHOM to share these tagged sets with and for WHAT purposes.

4. (U//FOUO) No, however, USCCI 5000-05, Records Management Program, is currently in Draft waiting for the USCC Chief of Staff (CoS) signature.
5. (U//FOUO) Yes/No. The J2 Directorate adheres to the standards set forth by both USCYBERCOM, Joint Staff, and DIA. We have created templates and have explained to Records Technicians the importance of tagging documents. However, enforcing metadata tagging is still a process which we are working through.
6. (U//FOUO) No, For our web-authorized platform, SharePoint, most lists have metadata associated with the items. However, the metadata is not required nor standardized.
7. (U//FOUO) Yes/No. Not for information sharing. For records management. Sharing within the organization is limited by functionally or organizationally restricted shared drives or SharePoint portal sites.
8. (U//FOUO) No, the Cyber functions within the Agency configure permissions based on the index and infrastructure. The Agency is migrating to an open source schema that will facilitate the future need for information sharing based on tagging.
9. (U//FOUO) No. There are some functional areas that are more advanced in using metadata tagging than other areas. Metadata tagging is currently a manual task and is tied to human behaviors. For instance, the Air Force has more than 5300 records retention codes. Having two different people tag the record with the same retention code is highly unlikely. The Air Force Medical Service, since 2003, has deployed data catalogs for the specific purpose of increasing data transparency, controlling access and distribution of data and identifying documents of record. Other functional areas have been very slow in adopting data catalogs but we are making progress.
10. (U//FOUO) Yes, through technical measures enforced by SharePoint.

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Q4. Do the tagging standards used address these 5 metadata tagging categories, check all that apply

Organization	CIA	JAIC	Cyber Security	USSOUTHCOM	USCYBERCOM J010	USCYBERCOM J2	USCYBERCOM J3	USCYBERCOM J070	USCENTCOM J2	USCENTCOM JIDC	NGA	Air Force	USEUCOM
Discovery, also known as Query	Y	N	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y
Access Control	Y	N	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y
Correlation and Analysis	Y	N	Y	Y	Y	Y		U	N	Y	Y	Y	Y
Monitoring, also known as Audit	Y	N	Y	Y	Y	Y		Y	N	N	Y	Y	Y
Records Management	Y	N	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y

Q5. What data tagging standards and/or specifications are being used to share information within your Organization or between your organization and other Department/Agencies:

- (U) The CIA Attributes Governance Council determines the ontology and typing of official attributes. Here are some of the authoritative sources for standards:
 - (U) IC CIO Technical Specifications and XML Standards:
https://intellipedia.intelink.ic.gov/wiki/IC_XML_Standards to include PUBS, ISM, NTK
 - (U) Agency XML Standards – for customizations to the IC Formats to handle local requirements
 - (U) Multiple standards are defined for serialization of those attributes, including their association with data.
 - (U) MINT: <https://intellipedia.intelink.ic.gov/wiki/MINT>

(U) The primary standards used that span multiple systems are concentrated in access control and classification markings. The IdAM standards ensure data is consistently protected across multiple systems, but continue to be an implementation challenge. The classification marking standards are most consistently used with formal communications, cables and intelligence documents.

- (U//FOUO) For Information warfare, we use the DST-3 standard and have pushed a cyber-data format to help cyber communities share data. From a dataset management process, we used the terms defined by the Army Cyber Ontology
- (U) Utilizing the applicable tags for information sharing. Re-evaluating tags over time with assistance from the author of the content.
- (U//FOUO) USCCI 5000-05, Records Management Program, is currently in Draft waiting for the USCC Chief of Staff signature.
- (U//FOUO) Metadata is two-fold; based upon the Division File Plan created – series, bucket numbers, and single word descriptors are used for tagging. Key words specific/associated with

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the J2 including division names, and special classification are also used to tag documents created.

6. (U//FOUO) USCYBERCOM J0 developed a document sharing standard operating procedures (SOP) that includes a metatagging baseline for SharePoint. This SOP completed Action Officer staffing and will be presented for Command Chief of Staff signature.
7. (U//FOUO) MDSS-based metatagging and IC PUBS schema is used to ensure intelligence products are registered in some national databases for discovery in accordance with ICD 501.
8. (U//FOUO) Our environment employs the Intelligence Communities Trusted Data Format (IC-TDF) for data tagging. This standard includes:
 - (Required) Enterprise Data Header > Access Rights & Handling > Information Security Markings XML DES for Access Control (DISR MANDATED for Access Management)
 - (Optional) Information Resource Metadata (IRM) XML DES for Discovery and limited Correlation and Analysis (DISR MANDATED for Discovery Metadata)
 - (Optional) Electronic Records Management XML DES for records management (DISR MANDATED for Records Management)
 - (Optional) Mission specific fields for further Correlation and Analysis
 - (Related) There is a related Audit XML DES that JIDC has not implemented (DISR MANDATED for Cybersecurity Audit)

The TDF also includes the ability to use the now deprecated Department of Defense Discovery Metadata Specification (DDMS) 5.0.

9. NGA has been adopting the Elastic Common Schema (ECS) which is one for one aligned to the Cyber Data Framework (CDF). This schema will be the required standard for all Cyber Data within the Agency and all Cyber Data sets will be migrated to this standard no later than Q4 FY21.
10. For structured data the Air Force is using Informatica to identify metadata tags for automatic metadata TAGGING; For unstructured data the Air Force uses authoritative sources found in the Air Force ePubs library.
11. SharePoint capabilities such as catalogs of Enterprise Keywords, Custom Keywords, and other specifications: technical system requirement UNCLASS/CUI/CONFIDENTIAL/SECRET, etc. tags, or CCMD unique fields/content such as CCP LOE or Battle Rhythm Event.

Q6. Describe any impediments to using DoD/IC or other organizationally mandated standards and specifications.

1. (U) Ensure that tags and schema/structure are clearly defined, so they can be accurately and consistently applied. We also recommend avoiding IC- and DoD-specific terminology where possible, instead leaning on existing best practices and standards, facilitate consistent applications across agencies despite our differing missions and institutional cultures.

(U) The logical places to implement and enforce tagging standards in a more comprehensive way are often disconnected from the analytic or exploitation platforms where data is queried and used by analysts. There is a dichotomy between specialized tools accomplishing their

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somewhat narrowly funded and narrowly scoped missions (i.e. identity intelligence or weapons and counterproliferation) and the more universal goals of generalized discovery and correlation. Specialized tools are not in a position to push tagging requirements upstream, nor are there widely applicable tagging standards they would particularly advocate, and consequently all data is re-ingested or reprocessed when it gets to them. Formalized intelligence disciplines like SIGINT and IMINT are better equipped with tagging and formatting standards, but their successes are not all attributable to mandates and policies requiring the use of those standards.

2. I think the question is to define interlocking standards and not domain specific ones. Each domain has its own set of stake holders which are resistant to being forced into something from the outside. They first need to understand the need for sharing information (DoD does not do this well), then what they could do with data they can get from others, and then they will be more open to it.

Also data sharing hampered by security concerns. So a SCG for shared information would be good.

3. Personnel should have a better understanding as to where they should store their data. Most personnel use their personal shared drives or their desktops to store information. We have to get into the culture of document and content management which our KM office is working diligently to implement and enforce.
4. Data tagging would make documents more discoverable and would decrease redundancy in files created and allow the J2 workforce the ability to quickly provide support to mission and/or FOIA request.
5. Although we have technical measures in place, the time and effort for users to tag information, and appreciate their ROI for that time and effort, is under appreciated. Users have a myriad of choices on where to store their data, and may tend to store it where it is fastest and easiest for them. Given the option to save on their personal drive, or shared network drive, it is faster and easier because it does not require tagging. USEUCOM supports data management and modernization efforts for maturing data tagging across the organization, and the additional value AI/ML applied to large data sets can provide.
6. Impediments are difficulty in deploying data catalogs into production environments to address the same five categories from Question 4:
 - Discovery, also known as Query
 - Access Control
 - Correlation and Analysis
 - Monitoring, also known as Audit
 - Records Management
7. Vendor alignment and support tends to be challenging. Most standards cover higher level requirements but tend to fail operational level use cases at the organization level. Fiscal impacts are rarely addressed resulting in operational issues finding resources to meet policy.
8. USCYBERCOM has found that mandating standards and specifications is not sufficient to allow for interoperability of tagged data because organizations implement the standards differently. The IC has a central working group under the IC CDO Council to craft definitions and field-value structures to manage data effectively that are then required to be used to index and categorize

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data to enable interoperability between capabilities. DoD does not have a matching organization with the authority to mandate how data should be formatted within different standards. For example, 144.51.12.18 and 2419264530 are two different ways to represent an IP address. At a singular scale, conversion is not challenging, but with the scale of very large data sets, continuously converting between different formats is computationally expensive and many resources are expended converting data fields so queries across different data sets are able to properly function. USCYBERCOM is exploring current policy and legal space to determine if CDRUSCYBERCOM has the operational authority to order specific implementations of data standards in order to enable the Command to complete its assigned mission to defend the DODIN.

9. List of issues:

- The specifications are difficult to apply to standard apps (MS Office, SharePoint, etc.) and frequent versioning compounds the issue.
- Indexed search for tagged items is inadequate which lessens the emphasis on its need.
- Many users/producers don't understand tagging, its function or need.

10. The largest impediment to using mandated standards is the Return On Investment (ROI) to the user. Unless the user has a compelling use-case for tagging their data at point of creation then standards implementations – no matter how “mandated” – will be ultimately futile. If anyone is going tag data ISO of AI/ML (discovery, correlation, analysis) then they need to go ahead and include a specification for access control – and vice versa – this will save untold amounts of time on the back end as both will be required for success. This is especially true of DoD's Joint All-Domain Command and Control (JADC2) concept.

Appendix B. Policy Review Documents

Federal Policy

1. Intelligence Reform and Terrorism Prevention Act of 2004 (IRTPA) (Pub. L. No. 108-458), <<https://www.dni.gov/index.php/ic-legal-reference-book/intelligence-reform-and-terrorism-prevention-act-of-2004>>
2. Federal Records Act (P.L. 90-620), as amended, (44 U.S.C. §3301), <<https://www.govinfo.gov/app/details/USCODE-2008-title44/USCODE-2008-title44-chap33-sec3301>>
3. National Security Directive (NSD) 42, National Policy for the Security of National Security Telecommunications and Information Systems, 5 July 1990, <<https://fas.org/irp/offdocs/nsd/nsd42.pdf>>
4. **“National Strategy for Information Sharing and Safeguarding”, The White House, December 2012**, <https://www.dhs.gov/sites/default/files/publications/15_1026_NSI_National-Strategy-Information-Sharing-Safeguarding.pdf >
5. *Strategic Implementation Plan* for the National Strategy for Information Sharing and Safeguarding, Dec 2013, <https://www.hsdl.org/?view&did=748441>
6. **Program Manager, Information Sharing Environment** Data Tagging Functional Requirements, Dec 2014, <https://www.dni.gov/files/ISE/documents/DocumentLibrary/PO3-Data-Tagging-Functional-Requirements.pdf>
7. **Program Manager, Information Sharing Environment** Data Aggregation Reference Architecture (DARA), Dec 2014, https://www.dni.gov/files/ISE/documents/DocumentLibrary/DARA_v1.pdf
8. Office of Management and Budget (OMB) Circular A-130, *Managing Information as a Strategic Resource*, July 2016 < <https://www.cio.gov/policies-and-priorities/circular-a-130/>>

Committee for National Security Systems (CNSS)

All CNSS Documents listed are available from the CNSS library at CNSS.gov. Some of the documents are access restricted.

9. CNSS-007-07, Policy Development Guidance to Address Information Sharing for National Security Systems, 2007.
10. CNSSP-24, Policy on Assured Information Sharing (AIS) for National Security Systems (NSS), May 2010,
11. CNSSD-507, National Directive for Identity, Credential, and Access Management, July 2020.
12. Committee on National Security Systems (CNSS) Instruction 1253, Security Categorization and Security Control Selection for National Security Systems, March 2014

Appendix B. Policy Review Documents

National Institute of Standards

13. NIST Special Publication 800-63-3 Digital Identity Guidelines
<https://www.nist.gov/topics/identity-access-management/roadmap-nist-special-publication-800-63-3-digital-identity>
14. NIST Special Publication 800-162, Guide to ABAC Definition and Considerations,
<https://doi.org/10.6028/NIST.SP.800-162>
15. NIST Special Publication 800-207, Zero Trust Architecture
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-207.pdf>

Intelligence Community

Policy Documents

Access to ICS, ICPG, and ICD documents and memorandums can be obtained through
<https://intelshare.intelink.ic.gov/sites/policvandstrategv/Pages/IC-Policv.aspx>

16. ICS 500-20, IC Enterprise Standards Compliance
17. ICS 204-01, NIPF Tagging of Disseminated Intelligence
18. ICD 206, Sourcing Requirements for Disseminated Analytic Products
19. ICD 208, Maximizing the Utility of Analytic Products
20. ICD 209, Tearline Production and Dissemination
21. ICD 302, Document and Media Exploitation
22. ICPG 710.1, Dissemination Controls -Originator Control (ORCON)
23. ICPG 710.2/403.5, Dissemination Controls -Foreign Disclosure and Release Markings
24. ICD 501, Discovery and Dissemination or Retrieval of Information within the Intelligence Community
25. ICD 403, Foreign Disclosure and Release (FDR) of Classified National Intelligence
26. ICPG 403.1, Criteria for FDR of Classified National Intelligence
27. ICPG 403.5/710.2, Dissemination Controls -Foreign Disclosure and Release Markings
28. ICS 500-21, Tagging of Intelligence and Intelligence-Related Information
29. IC CIO 2018-081, Improving IC ICAM to Achieve Greater Effectiveness

Department of Defense

30. DoD Instruction (DoDI) 8320.02, Sharing Data, Information, and Information Technology (IT) Services in the DoD
31. DoDI 8320.03, Unique Identification (UID) Standards for Supporting the DoD Information Enterprise
<https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/832003p.pdf?ver=2019-04-24-141927-723>
32. DoDI 8320.07, Implementing the Sharing of Data, Information, and Information Technology (IT) Services in the DoD
33. DoD Instruction 5200.48 Controlled Unclassified Information
<https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520046p.pdf?ver=2018-05-04-092730-193>

Appendix C. Metadata Standards linked to Functional Requirement and Policy

The following chart lists capabilities from the *2012 National Strategy for Information Sharing and Safeguarding* Priority Objective 3, Data Tagging Functional Requirements, linked to DoD and IC policy and the specific metadata specification, as listed as of Feb 2021 in the DoD Information Technology Standards Registry (DISR).

Resource Attribute	Policy	Implementation	DISR Guidance
Unique Identifier	ICS 500-21 / DoDI 8320.03	Community Defined	Mandated
Discovery Metadata	ICS 500-21 / DoDI 8320.02	Information Resource Metadata (IRM.XML) or future Content Discovery and Retrieval (CDR) compliant specification	Mandated
Access Management	ICS 500-21 / DoDI 8320.02	Information Security Markings (ISM.XML)	Mandated
	DoDI 8520.03	DoD Enterprise Identity Attribute Service (EIAS)	Mandated
	ICS 500-30	IC Attribute Services, Authorization Attribute Set (UAAS)	Mandated
Cybersecurity Audit	ICS 500-27 / DoDI 8530.01	XML Data Encoding Specification for Enterprise Audit Exchange (AUDIT.XML)	Mandated
Records Management	ICD 501 / DoDI 5015.02	XML Data Encoding Specification for Electronic Records Management (ERM.XML)	DoD Emerging IC Mandated
Correlation and Analysis	ICS 500-21 / DoDI 8320.02	Various, see specific Mission Architecture	

Appendix C. Metadata Standards linked to Functional Requirement and Policy

The following is the complete list of technical specifications that are applicable to *2012 National Strategy for Information Sharing and Safeguarding* Priority Objective 3 data functions. Access to the Technical Specifications can be obtained through

<https://intelshare.intelink.ic.gov/sites/odni/cio/techspecs/Documents/CDR/ATOM/default.aspx>

1. Atom Result Set (v2.0), Atom Data Encoding Specification for Content Discovery and Retrieval Result Sets, Version 2 (10 Apr 13) [Scheduled for 13-2 Baseline for ER2 and DISR]
2. AUDIT.XML (v2019-MAR), Enterprise Audit Exchange (AUDIT.XML); Version 2019-MAR (8 Mar 2019), [19-2 Baseline]
3. AUTHCAT.CES (v2018-APR), Authority Categories (AUTHCAT.XML), Version 2018-APR
4. BOE.XML (v2018-APR), Body of Evidence (BOE.XML), Version 2018-APR (20 Apr 2018)
5. BRK-SRCH (REST & SOAP) (v2.0) , REST & SOAP Interface Encoding Specification for Content Discovery and Retrieval Brokered Search (CDR-RB & CDR-SB), Version 2 (6 Sep 13), [14-1Baseline for ER2 and DISR]
6. CDR (REST & SOAP) Deliver (v2) , IC/DoD REST Interface Encoding Specification for CDR Deliver, Version 2 (14 Mar 14) [14-2 Baseline]
7. CDR (REST & SOAP) Describe (v1.0), IC/DoD REST Interface Encoding Specification for CDR Describe, Version 1(14 Jan 13), [13-1Baseline]
8. IC/DoD SOAP Interface Encoding Specification for CDR Describe, Version 1 (14 Jan 13), [13-1Baseline]
9. CDR (REST & SOAP) Manage (v1), REST & SOAP for CDR: Manage, Version 1 (14 Mar 14) [14-2 Baseline]
10. CDR (REST & SOAP) QueryMgmt (v2), REST & SOAP For CDR Query Management, Version 2 (14 Mar 14) [14-2 Baseline]
11. CDR (REST & SOAP) Retrieve (v2.0), IC/DoD REST Interface Encoding Specification for CDR Retrieve, Version 2 (3 Oct 12)
12. IC/DoD SOAP Encoding Interface Specification for CDR Retrieve, Version 2 (3 Oct 12)
13. CDR (REST & SOAP) Search (v3.0), IC/DoD REST Interface Encoding Specification for CDR Search, Version 3 (3 Oct 12)
14. IC-DoD SOAP Interface Encoding Specification for CDR Search, Version 3 (3 Oct 12)
15. CDR RA (v2), IC/DoD Content Discovery and Retrieval Reference Architecture, Version 2 (10 Apr 13)
16. CDR.SF (v2), IC/DoD Content Discovery & Retrieval Specification Framework, Version 2 (10 Apr 13)
17. CEM.XML (v2018-APR), Contextual Entity Markup (CEM.XML), Version 2018-APR (20 Apr 2018), [18-3 Baseline]
18. CSR.XML (v2017-MAY), Community Shared Resources (CSR.XML); Version 2019-MAR (8 Mar 2019), [19-3 Baseline]
19. CSR-AUDIT.XML (v2019-MAR), Community Shared Resources (CSR.XML); Version 2019-MAR (8 Mar 2019), [19-3 Baseline]
20. DED.XML (v2019-MAR?), Data Element Definition (DED.XML), Version 2019-MAR (8 Mar 2019)
21. DOMEX.XML (v2015-AUG), Document and Media Exploitation (DOMEX.XML), Version 2015Aug (13 Aug 2015), [15-2 Baseline]
22. ERM.XML (v2016-JUL?), Electronic Records Management (ERM.XML), Version 2016-JUL (29 Jul 2016), [16-3 Baseline]

Appendix C. Metadata Standards linked to Functional Requirement and Policy

23. GENC.CES (v2019-SEP), Geopolitical Entities, Names, and Codes (GENC.XML); Version 2019-SEP (6 Sep 2019), [20-1Baseline]
24. IC.ADD (v2), Intelligence Community Abstract Data Definition
25. IC.ADD-ERM (v2014-DEC?), Abstract Data Definition for Electronic Records Management
26. IC-DOC.XML (v2018-APR?), Intelligence Community Docbook (IC-Docbook.XML), Version2018-APR (20 Apr 2018), [18-3 Baseline]
27. IC-EDH.XML (v2019-MAR), Enterprise Data Header (IC-EDH.XML); Version 2019-MAR (8 Mar2019), [19-2 Baseline]
28. IC-ID.XML (v1), Intelligence Community Identifier (IC-ID.XML), Version 1 (10 Apr13), [13-2 Baseline]
29. IC-SF.XML (v2019-MAR), IC Specification Framework(IC-SF.XML); Version 2019-MAR (8Mar 2019)
30. INTDIS.CES (v2017-JUL), Intelligence Discipline (INTDIS.XML); Version 2017-Jul (21Jul2017) [17-3 Baseline]
31. IRM.XML (v2019-MAR), Information Resource Metadata (IRM.XML); Version 2019-MAR(8 Mar 2019), [19-2 Baseline]
32. IRM-DDMS-SRCH.XML (v2), DoD Discovery Metadata Specification (IRM-DDMS-SRCHXML) Version 2 (6 Sept 13), [14-1Baseline: Info Guidance for ER2 and DISR]
33. ISM.ACES (v2019-SEP), Access Control Encoding Specification for Information Security Markings, Version 2019-SEP (6 Sep 2019) [20-1Baseline]
34. ISM.XML (v2019-MAR-r2019-SEP), Information Security Marking Metadata (ISM.XML), Version 2019MARr2019-SEP (6 Sep 2019) [20-1 Baseline]
35. ISMCAT.CES (v2019-SEP), XML CVE Encoding Specification for ISM Country Codes and Tetragraphs (ISMCAT.XML); Version 2019-SEP (6 Sep 2019)
36. ITS-MS.XML (v2015-FEBBr2018-JUL), Information Transport Service Messaging Service (ITS-MS.XML); Version 2015-FEBBr2018-JUL (31Jul 18), [18-3 Baseline]
37. ITS-OM.XML (v2015-AUG), Information Transport Service Organizational Messaging (ITS-OM.XML), Version 2015-AUG (13 Aug 2015), [15-2 Baseline]
38. Keyword Query (v2.0), IC/DOD Keyword Query Language Specification, Version 2 (3 Oct 12)
39. LIC.CES (v2015-AUG), License (LIC.XML), Version 2015-Aug (13 Aug 2015), [15-2 Baseline]
40. MAC.XML (v2016-SEP), Multi Audience Collection (MAC.XML); Version 2016-SEP (9 Sep 16), [16-3 Baseline]
41. MIME.CES (v2016-SEP), Media Type Controlled Vocabulary (MIME.XML); Version 2016-SEP (9 Sep 16), [16-3 Baseline]
42. MN.CES (v2017-MAYr2019-MAR), Mission Need (MN.XML); Version 2017-MAYr2019-MAR (8 Mar 2019), [19-2 Baseline]
43. MNT.TES (v2019-MAR), XML Taxonomy Encoding Specification for Mission Need Taxonomy (MNT.XML); Version 2019-MAR (8 Mar 2019), [19-2 Baseline]
44. PM.CES (v2019-MAR), Production Metrics (PM.XML); Version 2019-MAR (8 Mar 2019), [19-2 Baseline]
45. PMA.XML (v2019-MARr2019-SEP), Production Metrics Assertion (PMA.XML), Version 2019 MARr2019-SEP (6 Sep 2019), [20-1Baseline]
46. PUBS.XML (v2018-APR), PUBS Convenience 2019-SEP, Version 2019-NOV (6 NOV 2019) Intelligence Publications (PUBS.XML), Version 2018-APR (20 Apr 2018), [18-3 Baseline]
47. REVRECALL.XML (v2014-DECr2018-APR), Revision Recall (RevRecall.XML); Version 2014-DECr2018-APR (20 Apr 2018), [18-3 Baseline]

Appendix C. Metadata Standards linked to Functional Requirement and Policy

48. RR-ID (v1), End-to-End Identity Propagation (RR-ID); Version 1 (17 Jul 12) [12-3 Baseline for ER2 and DISR]
49. RR-SM (v2?), Security Markings (RR-SM); Version 2 (14 Jan 13), [13-1Baseline for ER2 and DISR]
50. SRC.XML (v2015-AUG), SRC (SRC_2015-AUG.XML); Version 2015-Aug (13 Aug 2015), [15-2 Baseline]
51. TDF.XML (v2019-MAR), Trusted Data Format (IC-TDF.XML); Version 2019-MAR (8 Mar 2019), [19-2 Baseline]
52. USAgency.CES (v2017-MAR-r2018-FEB), XML CVE Encoding Specification for US Agency Acronyms (USAgency.CES); Version 2017-MARr2018-FEB, (16 Feb 18), [18-2 Baseline]
53. VIRT.XML (v2015-AUGr2017-JUL), Virtual Coverage (VIRT.XML), Version 2015-AUGr2017-Jul (21 Jul 2017), [17-3 Baseline]
54. WSS-HLG.XML (v1), High Level Guidance for Web Service Security (WSS-HLG.XML); Version 1 (10 April 13) [13-2 Baseline for ER2 and DISR]
55. WSS-SIGENC.XML (v1), WSS XML Signature & XML Encryption (WSS-SIGENC.XML); Version 1 (14 Mar 14), [14-2 Baseline]
56. WSS-TS.XML (v2014-DEC), Web Security Standards Guidance for Token Services (WSS-TS.XML); Version 2014-DEC (4 Dec 2014), [15-1Baseline for ER2 and DISR]